

YES, BUT CHOOSE THE RIGHT LAB!

 knowledgeiswatt.substack.com/p/55-are-metabolic-carts-for-vo2max

Knowledgeiswatt

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Photo by [VO2 Master](#) on [Unsplash](#)

VO2max, ventilatory thresholds and fuel utilisation (how many grams of carbs and fats you use at each intensity) can be interesting parameters to know.

- VO2max (maximal oxygen consumption), that is the maximum oxygen consumption rate, can give an idea of the aerobic engine of the athlete.
- Ventilatory thresholds can be used to divide training zones to plan workouts. They are an alternative to other methods more used on the field, like lactate and CP/FTP. For example, when using a 5/7 zones model, first ventilatory threshold should divide zone 2-zone 3 and the second ventilatory threshold should divide zone 4-zone 5.
- Fuel utilisation parameters can give you an idea of how much you can be durable. For example if your peak fat oxidation and fatmax (read more [here](#) about these parameters) are high, you have more chance to be more durable (resistance to fatigue during long rides).

To directly measure VO2max, ventilatory thresholds and fuel utilisation parameters, you need a *metabolic cart*, that is a machine that measures gases expired through a mask and can then calculate oxygen consumption, carbon dioxide produced, ventilation rate and breath frequency. After collecting data you can define:

- **VO2max**; the highest rate of oxygen consumption (ml/min/kg) recorded during an incremental maximal test.
- first ventilatory threshold: the first disproportionate increase in ventilation rate compared to oxygen consumption (caused by an increase in ventilatory drive stimulated by the increase of blood carbon dioxide derived by blood bicarbonate buffering the H⁺ produced by working muscles).
- second ventilatory threshold: the disproportionate increase in ventilation rate compared to carbon dioxide production, that is caused by the abrupt increase in blood H⁺ ions, as at this exercise intensity the blood bicarbonate buffering system cannot compensate anymore for the increase of acidosis (read H⁺).
- Fuel utilisation parameters: applying stoichiometric equations to oxygen consumption and carbon dioxide produced. You can download the excel file to do this [here](#).

These are the information you can have after a test with a metabolic cart. The price for such a test is usually between 150 and 400 euros.

As you are spending money, you would like to have accurate test results, but this is not always the case. There are several metabolic cart models on the market, but not all are enough accurate.

Which metabolic carts on the market are more accurate?

A hot off the press study published in 2024 by Van Hooren and Colleagues (Maastricht University, Netherland) on *Scandinavian Journal of Medicine & Science in Sports* tried to answer this question.

Answer this question can help to:

- if you are an athlete, which lab paying to do the test (based on the metabolic cart they have)
- If you are a coach or an exercise physiologist and want to buy a metabolic cart to test athletes, this KisW research note can help you to select a good brand and avoiding wasting your saves in an instrument which can't record accurate data.

WHAT DID THEY DO?

- They took 15 popular and commercially available metabolic carts and compared them with a state-of-the-art metabolic simulator consisting of a breathing simulator combined with a gas-infusion system.
- In this way they were able to verify if each metabolic cart is accurate or not.
- The verifications were made simulating different exercise intensities and calculating the errors between each metabolic cart and metabolic simulator.

HERE THE RESULTS!

Here below the overall error for each metabolic cart considered as the average for the following parameters: ventilation rate, breath frequency, oxygen consumption, carbon dioxide produced and respiratory exchange ratio (which can help to calculate carbs and fat used):

- **Vyntus CPX: 0.56%**
- Oxycon PRO B X B: -1.19%
- Oxycon PRO MC: -3.13%
- Omnicol V6: -2.57%
- Ergostik: -1.23%
- MetaLyzer 3B: +1.4%
- **MetaMax 3B: +0.39%**
- VO2masterPro: -5.82%
- PowerCube Ergo: +5.78%

- QUARK CPET: -2.15%
- Ultima CPX: -4.45%
- **Ergocard CPX Clinical: +0.72%**
- **Ergocard CPX PRO: -0.34%**
- K5: - 3.05%
- PNOE: +10.1%
- **Calibre: -0.63%**

Please note that, of course, higher the error (both positive + or negative -, worse the product!

PRACTICAL TAKEAWAYS:

Not all the metabolic cart on the market are accurate enough to be used.

GABRIELE'S COMMENT:

- No much more to add. I would use only metabolic cart below 2% error.
- You can use these data to evaluate:
 - going or not in a given exercise physiology lab to spend money for your test (according to the metabolic cart they have). **If you can't find online which metabolic cart have the performance centre you are going to visit, I would suggest to mail them asking about this information. This can save you time and money.**
 - choosing the metabolic cart to buy if you are a coach or exercise physiologist.

Thanks for Reading! Spread with your Team!

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REFERENCES:

1. Van Hooren B, Souren T, Bongers BC. Accuracy of respiratory gas variables, substrate, and energy use from 15 CPET systems during simulated and human exercise. Scand J Med Sci Sports. 2024 Jan;34(1):e14490.



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